

AWS × VERCEL HACKATHON · TRACK 3 · MILLION-SCALE GLOBAL APP

PulseRooms

Real-time global trivia — built to absorb millions of concurrent answers.

Amazon DynamoDB · single-table design

PulseRooms

► pulserooms.vercel.app · github.com/manojmallick/pulserooms

THE PROBLEM

— Why this is hard

A live trivia round is a write firehose with a hot, real-time leaderboard read — globally.

A relational DB buckles under millions of concurrent answer-writes.

Naive DynamoDB sends every write to one partition → guaranteed hot-partition throttling.

THE SOLUTION

— How it works

1

Shard writes

Per-room leaderboard write-sharded across N partitions (GSI1) — no hot partition.

2

Scatter-gather

Reads fan out across shards and merge the global top-K. Season board on GSI2.

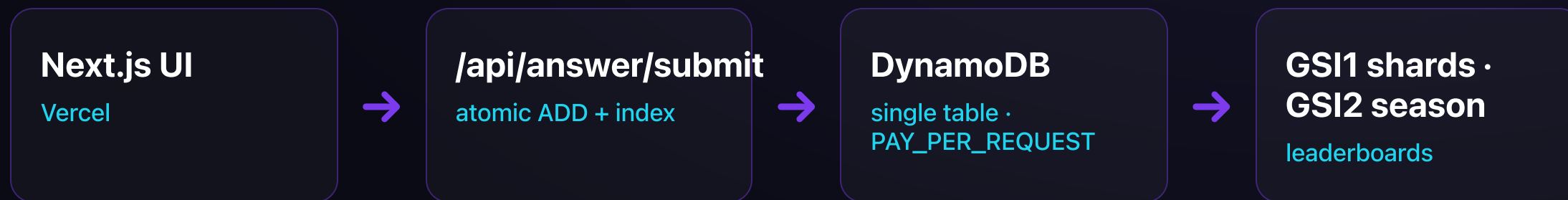
3

Score correctly

Atomic ADD score → index at the TRUE total; server-authoritative grading.

ARCHITECTURE

— End to end



WHY AMAZON DYNAMODB ·

Single-digit-ms latency at any scale, Streams for fanout, Global Tables for active-active — the textbook write-firehose profile.

TECH & DIFFERENTIATION

— Built with

Next.js 16

TypeScript

Amazon DynamoDB

AWS SDK v3

Global Tables

Vercel

WHY THIS DATABASE

Single-digit-ms latency at any scale, Streams for fanout, Global Tables for active-active — the textbook write-firehose profile.

THE EDGE

Write-sharded leaderboard + correct atomic scoring flow + a load-test proof harness. Profiles, streaks & history in the same table.

TRACTION

— Live & shippable

Live now with a global seasonal leaderboard. Architecture rigor over concept novelty.

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